

ROY CARLOUS CHRISTUDASS

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EDUCATION

Wright State University | Fairborn, OH

Aug 2024 – May 2026

Master of Science in Computer Science

Relevant Coursework: Algorithm Design and Analysis, Distributed Computing, Foundations of AI, Advanced Computer Networks

Sathyabama Institute of Science and Technology | Chennai, India

Aug 2018 – Jun 2022

Bachelor of Engineering in Electronics and Communication Engineering

SKILLS

Programming Languages: Java, Python, C#, C++, JavaScript, HTML/CSS

Cloud & DevOps: AWS (SQS, S3, DynamoDB), Docker, Docker Compose

Frameworks & Libraries: Spring Boot, FastAPI, React, SQLAlchemy, scikit-learn, Tree-sitter

Databases: PostgreSQL, DynamoDB, H2

Tools: Git, Maven, pytest, JUnit, JIRA, VersionVault, Swagger/OpenAPI, Ollama

Concepts: Distributed Systems, REST APIs, Message Queues, OOP Design Patterns, ML Pipelines, Static Analysis

PROFESSIONAL EXPERIENCE

Senior Software Engineer | **HCLTech** (Contractor to Teradyne) | Chennai, India

Jan 2024 – Aug 2024

- Developed and maintained high-performance C++ and C# instrument drivers for Teradyne's IGXL semiconductor test platform (Flex, UltraFlex, UltraFlex+ systems)
- Resolved 150+ software defects and acceptance test issues across multiple product lines, ensuring regression-free releases
- Introduced new language nodes for analog instruments within the IGXL environment, enhancing system capabilities for semiconductor test engineering

Software Engineer | **HCLTech** (Contractor to Teradyne) | Chennai, India

Aug 2022 – Dec 2023

- Extended complex driver architectures to implement new instrument capabilities for next-generation test hardware within an Agile C++ and C#/.NET engineering team
- Developed comprehensive unit tests and auto test suites, fixing numerous defective acceptance tests and aligning coverage with evolving IGXL requirements
- Managed source code integrity through VersionVault (ClearCase) with rigorous branching and merge strategies across Analog, Core, and Digital codebases

Graduate Engineer Trainee | **HCLTech** (Intern) | Chennai, India

Jan 2022 – Jul 2022

- Completed technical training in C++, C#/.NET, and OOP/OOD principles, gaining hands-on exposure to analog instrument driver architectures and semiconductor test equipment
- Collaborated with cross-functional teams to support IGXL platform workflows, contributing to status reporting and chip docking processes via JIRA-tracked delivery

PROJECTS

TaskForge – Distributed Report Generation Engine

[Live Demo](#) | [GitHub](#)

Java, Spring Boot, AWS (SQS, DynamoDB, S3), Docker, React, H2

- Built a distributed report generation system where requests are submitted via REST API, queued through SQS, processed by independent workers, and stored in S3 with presigned download URLs with TTL expiry
- Implemented fault-tolerance patterns including exponential backoff with jitter, dead letter queues, idempotency enforcement, graceful shutdown, and rate limiting (60 req/min per IP)
- Designed correlation ID tracing across the full pipeline, three report generators from real H2 business data, and an embedded React dashboard with live job status tracking

DiffLens – ML-Powered Code Review Engine

[Live Demo](#) | [GitHub](#)

Python, FastAPI, scikit-learn, Tree-sitter, PostgreSQL, React, Docker, Ollama

- Built a static analysis engine that evaluates Python and Java code for complexity, naming conventions, and bug risk patterns using Tree-sitter AST parsing and gradient boosting ML risk scoring
- Integrated GitHub webhooks to automatically analyze pull requests and post review comments with severity-ranked findings directly on PRs
- Implemented FastAPI backend with PostgreSQL review storage, React dashboard with data visualization, Docker Compose deployment, and a 14-file test suite

Portfolio Website – React, Vite, Tailwind CSS, Gemini 2.0 Flash API

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- Built a single-page portfolio with a cinematic multi-phase intro animation using HTML5 Canvas rendering and a phase state machine, with a real-time background particle system
- Integrated Google Gemini 2.0 Flash API with context-aware prompt injection, dual-key rotation for failover, client-side rate limiting (15 req/5 min), and in-memory response caching